

Cold Referee Report: Decidability of $\mathcal{ALCI}_{\text{RCC5}}$ Concept Satisfiability

Packet 2 of 3: Major Findings in Detail

Eighth review of the split-forest stack

July 9, 2026

Notation as in the manuscripts: P_t the live ports of bag t , Univ_T the universal sources of the tree T , $A_t = P_t \cup \text{Univ}_T$ the active set, N_t^* the coherent active completion required by AV1, $\rho(x, y)$ atomic RCC5 relations, Comp the (machine-verified) composition table, converses $\text{PP}^{-1} = \text{PPI}$, $\text{DR}^{-1} = \text{DR}$, $\text{PO}^{-1} = \text{PO}$. All composition values quoted below were checked by computation (Packet 3, Section D).

Finding F1 (critical). Theorem C has no proof in the effective stack, and the Section-6 monitor design is unsound as described

Locations. Round-14 Section 6 (automaton delta), Remark 6.1, Theorem 8.1, Corollary 8.2; round-14 Definition 2.2 (AV1–AV4); round-12 Section 5 with Appendices H, N (voided, see below); round-13 Remark 3.3 and Section 9 (withdrawn).

F1.1 The statement of Theorem C no longer matches its object

In round 12, a valid abstraction was a labelled tree over a finite alphabet Σ , and Theorem C asserted $L(\mathcal{A}_{C_0}) \neq \emptyset$ iff a valid abstraction with root C_0 exists; Lemmas 5.2/5.3 proved the two inclusions *for the round-12 validity clauses*. Those clauses have since been rewritten twice (round 13: D1–D5 and Λ ; round 14: AV1–AV6). Consequently Lemmas 5.2/5.3 as proved are void for the effective stack, and no analogue is restated or proved in either delta. Round 14 is candid about this: Section 6 is offered “so that the proof document is internally consistent . . . a review target rather than a mechanized certification,” and Remark 6.1 defers the “spawning discipline for pair and triple states.” As of round 14, the effective stack contains *no proof of any Theorem-C statement*. That alone suffices for the gap verdict, since Corollary 8.2 (decidability) is obtained by quoting exactly this missing theorem.

The deeper problem is that the missing statement is not even of the round-12 form. A valid AV abstraction is a pair (T, N^*) : a tree T over a finite alphabet *plus* an enrichment N^* assigning an atomic relation to every pair in $\text{Univ}_T \times \bigcup_t P_t$ and in $\text{Univ}_T \times \text{Univ}_T$, subject to AV1–AV4. Round 14 states explicitly that the enrichment is not part of any bag label (“Shadows are not listed as an unbounded part of a bag label; as in round 12, they are generated by alternating run copies”). So the automaton runs on the impoverished tree T , and the correct target statement is a projection:

$$L(\mathcal{A}) \neq \emptyset \iff \exists T \exists \text{enrichment } N^* \text{ such that } (T, N^*) \text{ is valid.}$$

Whether the projection language $\{T : \exists N^* (T, N^*) \text{ valid}\}$ is recognizable by a finite two-way alternating parity tree automaton is precisely the open question, and nothing in the stack addresses it. The label “bookkeeping” in Remark 6.1 presupposes the answer.

F1.2 The enrichment cannot ride the tape

The standard technique for existentially quantified run data — put the guess into the input alphabet, so that all run copies read the same values — is unavailable here, by the stack’s own accounting. The number of simultaneously active universal sources at a single bag is unbounded in the concept size: round-14 Corollary 5.2’s own table for the G2a tower lists $m - 1$ virtual shadows at the root-edge bag of the depth- m instance, growing without bound as m grows for a *fixed* concept scheme, and the round-13 G2 discussion establishes the same for a single fixed concept along an infinite branch. A per-node annotation carrying N_t^* rows for all active sources therefore has unbounded size, i.e., an infinite alphabet. This is not an incidental design choice that could be reverted: recording rows in labels is exactly the round-12 mechanism (Appendix H.2’s shadow states update their vectors “using the relation vector declared in the adjacent bag label”), and it is exactly what rounds 13–14 abandoned because unboundedly many sources’ rows cannot all be declared (see F6, Packet 3, for the resulting broken citation). The gap has migrated across three rounds — bounded labels with an unproved existence claim (round 12), bounded labels with a restricted row language that broke width (round 13), total data with no automaton (round 14) — and has not closed.

F1.3 Run-copy guessing fails soundness: a concrete configuration

The Section-6 sketch has three species of copies: single-shadow monitors $q_{\text{sh}}(\sigma)$ carrying a source’s row vector $\vec{r}$ to the current ports; pair monitors $q_{\text{pair}}(\sigma_1, e, \sigma_2)$ carrying two profiles and the mutual relation e ; triple monitors for three-source triangles. All are finite-state, since $|P_t| \leq K$; cardinality is not the issue. The issue is that in an alternating automaton, distinct copies resolve their nondeterminism independently: nothing in the model of computation forces two copies that each carry a guessed value for the *same* pair to carry the *same* value. The sketch’s checks are therefore per-copy checks, and per-copy checks do not compose to AV1. Concretely:

Claim. *There is a configuration in which every monitor described in Section 6 passes all of its checks, yet no coherent N^* exists, so the input is not a valid AV abstraction and must be rejected.*

Take two universal sources x, y introduced in disjoint subtrees, and two ports p, q live in two further, far-apart bags; suppose type-level constraints (universal formulas at the four occurrences, each direction checkable because each occurrence is live somewhere and the other member of the pair is virtually active there) pin the true rows to

$$\text{row}(x, p) = \text{PP}, \quad \text{row}(y, p) = \text{PPI}, \quad \text{row}(x, q) = \text{PP}, \quad \text{row}(y, q) = \text{DR}.$$

Coherence of any candidate N^* then forces the source–source value $e = \rho(x, y)$ twice over, incompatibly:

$$e \in \text{Comp}(\rho(x, p), \rho(p, y)) = \text{Comp}(\text{PP}, \text{PPI}^{-1}) = \text{Comp}(\text{PP}, \text{PP}) = \{\text{PP}\},$$

$$e \in \text{Comp}(\rho(x, q), \rho(q, y)) = \text{Comp}(\text{PP}, \text{DR}^{-1}) = \text{Comp}(\text{PP}, \text{DR}) = \{\text{DR}\},$$

and $\{\text{PP}\} \cap \{\text{DR}\} = \emptyset$ (machine-checked). So no coherent N^* exists: AV1 is unsatisfiable over these rows, and a sound decision procedure must reject. Now run the sketched monitors. The single monitors $q_{\text{sh}}(\sigma_x)$, $q_{\text{sh}}(\sigma_y)$ guess the true rows and pass every AV4 check (the types were chosen consistently pairwise). The pair monitor for (x, y) carries *its own* copies of the rows: it is free to guess, say, $\text{row}'(x, p) = \text{row}'(y, p) = \text{DR}$, $\text{row}'(x, q) = \text{row}'(y, q) = \text{DR}$, and $e = \text{DR}$; every triangle it checks closes, since $\text{DR} \in \text{Comp}(\text{DR}, \text{DR}) = \text{Rel}$. No state of the sketch compares the pair monitor’s row copies with the single monitors’ rows — they are different copies of an alternating run, and the run tree contains no channel between them. Every described check passes; the run accepts; validity fails. Hence any Theorem-C soundness lemma is false for the natural reading of Section 6.

Two aggravating design holes, both concrete instances of Remark 6.1’s deferred “bookkeeping”: (i) *duplicate pair monitors*. For sources introduced at incomparable nodes u, v , the symmetric discipline (each source’s traveler spawns a pair monitor at the other’s introduction) creates two monitors for the unordered pair $\{x, y\}$ with independently guessed e ; a tie-break needs a canonical choice function on incomparable node pairs, which the two-way run must be able to evaluate locally — not impossible, but nowhere designed. (ii) *Triple monitors re-guess pair values*. Closure of the source–source network on Univ_T genuinely requires three-source triangles: the triple $(\rho(x, y), \rho(y, z), \rho(x, z)) = (\text{PP}, \text{PP}, \text{DR})$ is not closed, since $\text{DR} \notin \text{Comp}(\text{PP}, \text{PP}) = \{\text{PP}\}$ (machine-checked), yet each of its three edges can separately participate in closed triangles with ports. So triple checks are necessary; but a triple monitor’s three e -guesses are, again, independent of the three pair monitors’ guesses. Identity of sources is not the problem — occurrences are identifiable by their unique introduction events (C1) — value consistency is.

F1.4 Why the three natural repairs are blocked

(a) *Tape annotation* is blocked by unboundedness (F1.2). (b) *Profile quotient* — carrying rows per profile (τ, Ω) rather than per source, so that per-node data is bounded — loses information that the semantics needs: in the G2a tower $a_1 \text{ PP } a_2 \text{ PP } \dots$, the occurrences a_1 and a_5 have identical profiles, yet against the port a_3 their rows are forced to differ, $\rho(a_1, a_3) = \text{PP}$ by $\text{Comp}(\text{PP}, \text{PP}) = \{\text{PP}\}$ versus $\rho(a_5, a_3) = \text{PPI}$ by $\text{Comp}(\text{PPI}, \text{PPI}) = \{\text{PPI}\}$ (machine-checked). A quotient that identifies them cannot state AV1. Any viable coalescing needs a proven congruence — a lemma saying which distinctions are irrelevant — which is exactly the missing mathematics. (c) *Canonical (deterministic) guessing* — making the guessed values a function of the input so that all copies recompute the same values — has no candidate rule: the extraction (Theorem 5.1) defines N^* by reading true relations off an arbitrary model, and different models of the same tree yield different coherent enrichments; a canonical rule would amount to a uniformization theorem for the enrichment, again unproved and nontrivial.

F1.5 What would discharge F1

Any of: (i) a *bounded-interface lemma*: only boundedly many sources’ rows are “relevant” at each bag, with a proved existence lemma completing the rest — i.e., a round-13-style existence argument compatible with vertical rows; then rows ride the tape. (ii) A *bounded-annotation scheme* via source coalescing with a proven congruence (repairing route (b)). (iii) *Abandoning the tree-automaton layer* in favour of a mosaic / type-elimination / quasimodel argument executed directly on the infinitary AV abstraction — note that Theorem 3.3 and Theorem 4.3 already

supply the hard semantic half, so the missing piece is a finite certificate notion with a decidable existence check. Until one of these is carried out, Corollary 8.2 is unsupported.

Finding F2 (major). Request-source liveness (D4) is load-bearing for truth but absent from the width accounting

Locations. Round-14 AV5, Theorem 4.3 (existential case), Theorem 5.1 (final width paragraph), Corollary 5.2; round-13 D4, E4', Section 11(G2); round-12 Appendix C.2 (thresholds), O.1.

The dependence chain. Under AV, the active set is $A_t = P_t \cup \text{Univ}_T$: *non-universal* sources have no active rows anywhere. The round-14 truth lemma discharges a deferred existential $\exists R.D \in \text{tp}(x)$ by a parity-fulfilled vertical request “with exact bag relation R ,” and its proof closes with: “In all cases the relevant relation is a live bag value, hence is extended by ρ_T .” For the relation $\rho(x, q) = R$ at the fulfilment bag to be a *live bag value*, the source x must be live (co-bagged) there — for non-universal x there is no other mechanism in the effective stack, since the AV coherence machinery covers universal sources only. That is exactly round-13’s D4 (request-source liveness), retained by AV5’s back-reference to “the round-12 and round-13 witness clauses.” So D4 is not optional hygiene; the truth lemma is false without it.

The gap. Keeping x live along its support prefix extends x ’s trace through every intermediate bag, adding one port to each. Round-13 Section 11(G2) named precisely two width-pressure mechanisms: “the D2 fallback” and “D4 liveness.” Round 14 eliminates the first — Corollary 5.2 is a genuine and verifiable improvement (Packet 3, S10) — and says nothing about the second: the width paragraph of Theorem 5.1 argues only that *universal-shadow propagation* adds no ports. No lemma in the effective stack bounds (i) the length of a support prefix, or (ii) the number of distinct pending sources simultaneously live in one bag. Both quantities must be absorbed by $K_A(C_0)$ for Theorem 5.1’s width claim, and through it the automaton’s alphabet and state bounds, to stand.

Why I could not break it — and why that is not a proof. I attempted to force unbounded simultaneous liveness with a fixed concept and failed, instructively. The sharpest probe (p4, Packet 3) is the strong-equality tower

$$C_0 = A \sqcap \forall \text{PP}.\neg A \sqcap \forall \text{PPI}.\neg A \sqcap \forall \text{PO}.\neg A \sqcap \forall \text{DR}.\neg A \sqcap \exists \text{PP} . (\exists \text{PP} . \top \sqcap \forall \text{PP} . (\exists \text{PP} . \top \sqcap \exists \text{PPI} . A)),$$

satisfiable (nested-chain model, machine-checked sanity in Packet 3), in which the four guards make the A -element w globally unique, and every element a_k of the forced infinite PP-tower must witness $\exists \text{PPI} . A$ *by w itself* — one shared witness for infinitely many far-apart existentials. If each a_k discharged its existential by a request travelling down to w with a_k kept live en route, bags near w would accumulate $a_0, \dots, a_k$ for every k : unbounded width. The bounded presentation instead keeps w live along the entire ray (one connected trace, width +1) and fulfils every request locally. The stack’s extraction plausibly does this — Theorem A’s (S2) inserts selected subpart support nodes — but nothing legislates it: no lemma says the presentation must, or always can, choose fulfilment layouts whose liveness cost is bounded. The probe shows the choice has teeth; a width proof must make it.

What would discharge F2. A bounded-delay normal form: every satisfiable concept has a Theorem-A presentation in which every vertical request is fulfilled within distance $f(|C_0|)$ of its source, or (as in p4) the witness’s trace, rather than the source’s, is extended; plus an explicit

term in the K_A accounting for live request sources, with the threshold machinery of round-12 Appendix C.2 as the likely engine. Round-12 O.1’s “fulfilled at the next upward support occurrence on every lap” suggests the authors believe this; believing is not proving.

Finding F3 (major). The width bound: contradictory provenance and an unclosed recurrence

Locations. Round-14 Section 1 (patch statement) and Theorem 5.1; round-13 Section 8, Definition of $K_A(C_0)$, Lemma 8.2; round-12 Lemma 3.6 (withdrawn) and its F4/M1 finding.

(a) Scope contradiction. Round-14 Section 1 declares the document “a replacement for round-13 Sections 3–8.” Round-13’s Section 8 is the width-accounting section: it defines witness-generated contexts, proves Lemma 8.2 (the recurrence $B_{r+1} \leq c(n)(1 + B_r)^2$ over ranks $r \leq n$), and defines $K_A(C_0) \leq B_n$. Round-14 Theorem 5.1 then asserts width “at most the Theorem A bound $K_A(C_0)$ ” — a quantity defined nowhere in round 14 and nowhere in round 12 (round-12’s own bound, Lemma 3.6, was found defective in the sixth review and withdrawn in favour of round-13 Section 8). As literally written, the effective stack therefore either (i) withdraws its only surviving width-bound proof while still quoting its conclusion, or (ii) mis-states its own patch scope. This review follows its instructions and reads Section 8 as surviving; either way an erratum is required, and the ambiguity is not cosmetic: the width bound feeds Theorem 5.1, the alphabet bound, and every finiteness claim downstream.

(b) The surviving proof does not close. Lemma 8.2’s recurrence counts, per generated port, one round of (g2) witness generation, (g3) incomparable-superpart insertion, and (g4) pairwise boundary insertion, with rank decreasing only at (g5) side-context descent. But (g2) witnesses are themselves “port[s] already generated” in the lemma’s own sense, hence eligible to spawn further (g2) generation *at the same rank*; likewise (g4) insertions are generated ports. The proof’s entire treatment of this regress is one sentence — every port has a finite generation chain terminating in (g1), “so the whole context is contained in the union counted by the recurrence” — which establishes that each chain is finite, not that same-rank chains have bounded length, which is what the recurrence needs: if (g2) chains at a fixed rank can have unbounded length, B_r as defined bounds nothing. The intended repair is visible between the lines: witness formulas have strictly smaller modal depth, so if rank is *defined* as remaining modal depth then (g2) also strictly decreases rank and the regress vanishes under a single decreasing measure. But that is not what the text defines or proves — the rank bookkeeping treats (g2) as rank-preserving — and the sixth review already rejected the round-12 predecessor of this lemma (F4/M1) for exactly this genre of under-specified termination. The width layer has now been unsound or under-proved in three consecutive rounds; it should be re-proved from a clean definition, not patched again.

What would discharge F3. An erratum fixing the patch scope; a re-statement of Lemma 8.2 with a single explicitly decreasing measure (remaining modal depth) covering (g2)–(g5), a proof that every generation step decreases it, and a closed-form bound; and, per F2, an added accounting term for request-source liveness.

End of Packet 2. Packet 3 contains the steps verified sound (S1–S10), minor findings (F4–F7), the adversarial probe log (p1–p7), the computational appendix, and recommendations (R1–R6).